

Aspect-Based Sentiment Analysis of Student Feedback Using Generative Large Language Models: A Literature Review

I. INTRODUCTION

Student feedback gives a rich but underutilized source of usable data for educational institutions. Every institute conducts service at the end of the semesters which consists sentiments which is applied to numerical ratings often losing the detailed, aspect specific opinions in free text comments. Aspect based sentimental analysis addresses this limitation by identifying specific opinion targets such as workload or instructor clarity and the sentiment associated with each aspect [1].

Interest in applying ABSA to education has grown alongside advancement in natural language processing. Although early, rule based and statistical methods achieved limited accuracy but needed heavy domain engineering which leads to the race of deep learning, specifically recurrent architectures with attention mechanisms greatly enhanced the aspect extraction from student feedback [2]. The pre-trained transformers like BERT and its derivatives have further improved the performance[3], [4]. Recently LLMs such as GPT-3.5, GPT-4, LLaMA and Gemini has unlocked new approaches like zero-shot and few-shot ABSA. This development reduces the need for extensive, time consuming data annotation which is a burden in educational applications[5], [6].

It remains unclear which Generative model provides optimal ABSA performance. This review examines (i) the evaluation of ABSA, (ii) its educational application, (iii) generative model comparisons, (iv) LLM capabilities and limitations.

II. EVOLUTION OF ABSA METHODOLOGIES

A. Traditional Machine learning Approaches

Early ABSA employed rule-based methods and traditional machine learning algorithms. Algorithms like Naïve Bayes, SVM, Decision Tree and Random Forest were applied on 100 student feedback records. Naïve Bayes achieved highest accuracy of 80% by outperforming Random Forest which was 76% accurate[7]. This disapproves the statement that “complex is always better”. The authors suggest that for short, focused text like student feedback assumption of Naïve Bayes captures the sentiment effectively without overfitting which is a major issue in complex models[8]. The foundational sentiment classification using bag of word representations[7]. The key limitation of traditional machine learning approaches is their dependence on manually constructed features and pre-defined classifications which generalise poorly across institutional contexts.

B. Deep Learning and Attention Mechanism

Recurrent neural networks with attention mechanisms represented a significant advance by learning contextual representations directly from raw text. Target dependent LSTM [9]that conditioned sentiment prediction on target aspects, achieving substantial improvement on benchmark datasets. [4] further proposed develop an attention-based LSTMs integrating aspect embeddings into attention models which enables the model to focus on aspect relevant words.

Bi-directional LSTM (BiLSTM) with attention models specifically for student’s course reviews reached the text from both forwards and backwards to capture full context and uses attention layer to understand the importance of each word. The model achieved accuracy of 92.24% which was more efficient than standard LSTMs about 88% accuracy and CNNs with 88% accuracy [10].

Deep learning advantage is automatic feature learning and eliminating manual engineering[11]. However, these models typically require large labelled datasets and substantial computational resources, potentially prohibitive for resource constraint institution [3].

C. Transformer-Based Models and Pre-Trained Language Models

BERT [4] introduced a paradigm shift by providing deeply contextualized token representations using bidirectional pre-training. For ABSA BERT’s sentence pair input was used by framing aspect sentiment classification as a question answering task [12], [13] focused on maximizing accuracy for the E-commerce fashion sector. They constructed a manually commented dataset of 3,500 Trust pilot reviews covering 14 business aspects like “shipping”, “return process” etc. The result of their comparative analysis which consisted of 5 transformer models which includes RoBERTa found to be a superior model with 92.53% F1 score outperforming BERT (90.12%) and DistilBERT (88.47%). DistilBERT which had 40% fewer parameters and faster version of BERT which offers high accuracy but it slightly lags behind RoBERTa in zero-shot accuracy. It was found that a fine tuned DistilBERT performs efficiently with 87% F1-score compared to larger models and cheaper to deploy

[14] which is an important consideration for resource constrained educational institutions. In educational settings BERT models applied to student feedbacks consistently achieved 85%-92% F1 score when fine-tuned on domain specific annotated data, though the dataset contained only 2,000 samples.

D. Large Language Models and Generative Approaches

The arrival of instruction tuned large language models refined through reinforcement learning from human feedback has fundamentally shifted how we approach aspect based sentimental analysis. Unlike traditional methods that requires huge amount of labelled data for specific training, these modern LLMs can identify specific features and sentiments just by following plain English instructions. Whether they are working zero shot or few shot, they offer a practical solution for resource limited educational settings.

Evaluation of few models like GPT-4, GPT-3.5 turbo and LLaMA-2 chat sizes from 7B, 13B to 70B[5]. They wanted to see how well these models could handle app reviews, a task very similar to analyzing student feedback. Even without any pre training on this specific dataset, every LLM they tested outperformed the traditional rule-based systems. GPT 4 stood out exceeding 23.6% ahead of the best rule-based baseline in zero shot test. When they gave it just 5 examples to learn from, its performance improved by 6% reaching 56.1% F1 score. For sentiment, GPT 4 was impressively accurate with 76% F1 score for positive sentiment but struggled more with neutral ones which was 45%. However, a fine-tuned RE-BERT model still held a 6% lead over GPT 4, showing that if you actually have the data to train a model from scratch, it's gold standard. A major disadvantage was that LLMs still performs better at zero-shot extraction i.e. identifying aspects which were never trained on[12]. However, these models can be easily be confused, hallucinated and require careful prompting strategies to remain reliable compared to the supervised models[11].

Other research backups the potential of these models with less informative areas. The Cantonese counselling transcripts[6] which is a tough, low resource environment similar to specialized academic subjects[3]. In this case GPT 3.5 and GPT 4 achieved accuracy of 92.1% and 95.3% respectively compared to models like LSTMs and SVMs. On a larger scale, [15] used a zero shot with DeBERTa-v3 and DistilBERT to scan over 17,000

tweets about AI. While they managed 38% accuracy rate for detecting aspects which proved a clear picture of how zero shot ABSA is convenient and scalable, but it's still has its limit when it comes to precision.

According to [5], GPT 4 is currently the top performer for aspect extractions and classification of sentiments. It is efficient because it has more parameters and follows complex instructions better which helps to think through a problem step by step. Google Gemini pro and Ultra have demonstrated strong performance on multi lingual NLP baseline. Gemini ultra outperforms GPT 4 on certain reasoning tests [16] but study on this is not yet published.

Finally, the LLaMA based models are great because we can run them through our own hardware which is a greater advantage for resource limited educational fields and ensures that the student feedbacks never leave the university's secure servers which makes sure that the data is protected. Studies show that larger LLaMA-2-70B and the newer LLaMA-3-8B are surprisingly good at learning from just a few examples [5], [17]. Under the right conditions, they can perform almost as well as GPT 4 making them a strong option for schools that want to keep total control of their data.

III.ABSA IN EDUCATIONAL CONTEXT

Using ABSA for educational data is lot harder than using it for regular reviews. Students feedback is unique as it's full of informal talk, soft complaints, hidden meanings which is a clear example of implicit aspect references. For example, if a student writes "I wish there were more practice problems", they are actually unhappy with assessment part of the course, even though they never use that word. On other hand, one single comment can be a mix of sentiments where a student might love the teacher but hate the textbook.

Most research in this area has looked at online MOOC courses[10]. However, [18] have pointed out that student comments use way more personal pronounce and hidden topics than the product reviews do. This means that AI models like BERT which are usually trained on restaurant or laptop reviews don't work well. They need a lot of extra training before they can accurately understand a classroom setting[19].

One of the biggest problems is that there isn't enough labelled data to train these models properly. Most

studies use fewer than 2,000 examples which forces researchers to depend on guess work or general AI models. This project is going to address this gap by hand labelling about 5000 comments, we are creating a much bigger and better resource for training AI than almost any study done before.

IV. RESEARCH GAPS AND FUTURE DIRECTIONS

- 1) **Dataset Scarcity:** Most educational ABSA studies use <2,000 samples[3], [5]. This scarcity causes a major limitation for model development and comparison.
- 2) **Implicit Aspect Extraction:** Current models require explicit aspects[11]. Students often express opinions without naming aspects which was fundamental challenge faced.
- 3) **Joint Aspect Extraction and Sentiment Classification:** Most educational studies assume pre-defined aspects. Extracting aspects manually takes a lot of labour and often results in missing values. Multi-tasking improves performance but not on evaluated educational data[14].
- 4) **Resource Efficiency:** A good model requires substantial computation resource[13]. Educational institutions often have limited infrastructural resources[3]. Therefore, need for the lightweight, efficient models which are

specifically optimised for educational ABSA is required.

- 5) **Cross-Institution Generalization:** Models trained on one institution may not transfer to others due to varying terminology, feedback systems and student demographics.
- 6) **Explainability:** Deep learning models functions as “black boxes”. Educational administrators need to understand why these certain sentiments were detected for actionable resource.
- 7) **Gemini for ABSA:** No published study evaluates Google Gemini for ABSA in educational settings[16].
- 8) **LLM Hallucination:** LLMs can confuse aspects and misclassify neutral sentiments as negative[5]. No study has assessed these failure modes specifically in educational feedback settings.

V. CONCLUSION

This review has documented the evolutions of ABSA from rule based and statistical ML through attention enhanced deep learning, pre trained transformers and the latest generation of LLMs. The literature reveals a clear trend that LLMs are increasingly competitive with fine-tuned transformers in zero and few shot settings, particularly where labelled data are less available.

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APPENDIX

Table I. Traditional ML and Deep Learning Studies

Ref.	Model	Aspect Method	Dataset	Key Result
[7]	Naïve Bayes and SVM	Pre-defined	Product reviews	Foundational Sentiment analysis benchmarks
[8]	Naïve Bayes and Random forest	Pre-defined	Student feedback	Naïve Bayes – 80% Random forest – 76%
[9]	Target dependent LSTM	Pre-defined	Restaurant or Laptop	Aspect dependent Sentimental analysis
[2]	Attention Based Aspect Embedding LSTM	Aspect embeds	SemEval	Attention based Sentiment Analysis
[10]	BiLSTM and Attention	Ontology and Named Entity Recognition	MOOC reviews	92.24% accuracy

Table II. Transformer and LLM Studies

Ref.	Model	Setting	Dataset	Key Result
[12]	BERT (Question-answering mode)	Fine tuned	SemEval 2014	State of the art aspect sentiment analysis
[13]	RoBERTa	Fine tuned	E-commerce	F1 score-92.53%
[20]	DistilBERT	Fine tuned	Customer feedback	F1 score-87%
[5]	GPT-4/GPT-3.5/LLaMA-2	Zero/5 shot	App reviews (1,000)	GPT 4, 5 shot F1 score- 56%
[6]	GPT 3.5/GPT 4	Zero shot	Counselling (6,000)	GPT 4 accuracy- 95.3%
[15]	DeBERTa and DistilBERT	Zero shot	Tweets (17,500)	Accuracy 38%
[16]	Gemini ultra	Zero shot	Massive multitask language understanding benchmarks	Exceeds GPT 4
[17]	LlaMA-2 chat	Few shot	General NLP task	Competitive GPT 4